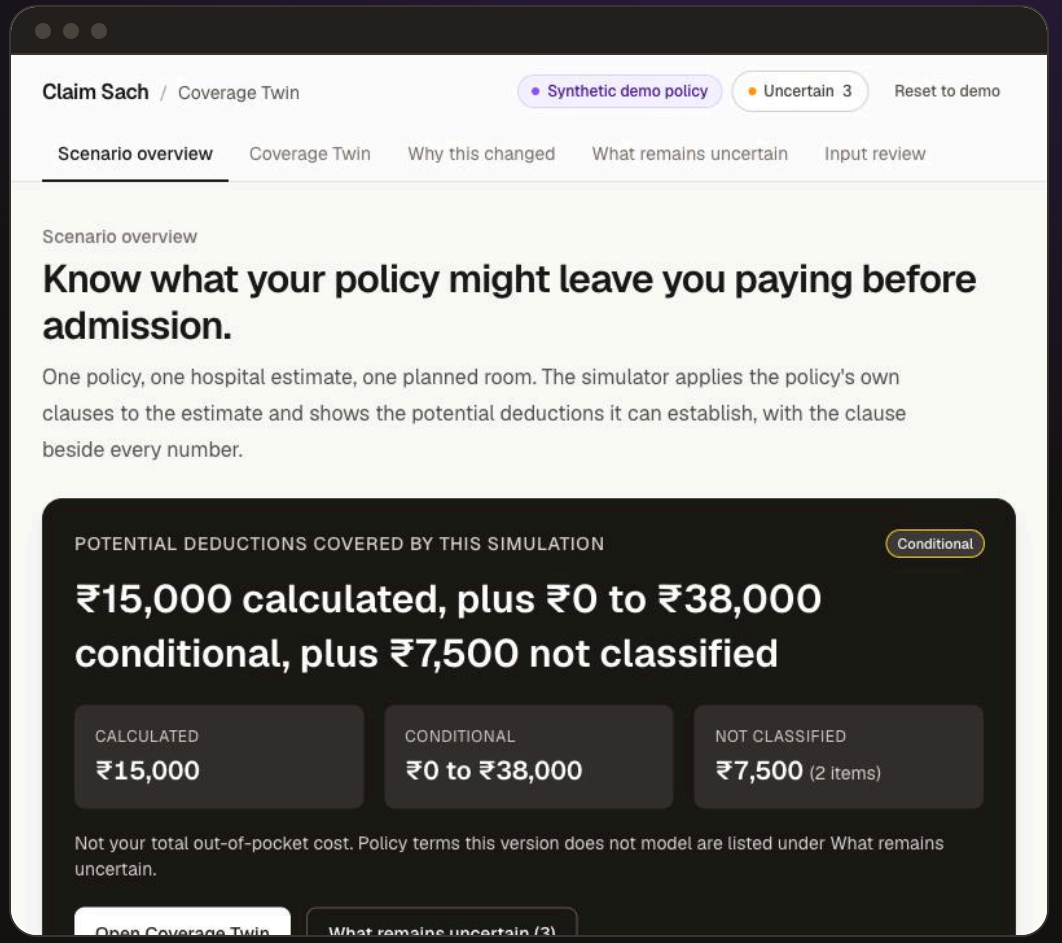
TRACK 2 · AI-POWERED FINANCIAL JOURNEYS

Claim Sach

Coverage Twin: know what your policy might leave you paying, *before* admission.

Pushkar Mondal



CONTENTS

Seven sections, one working product

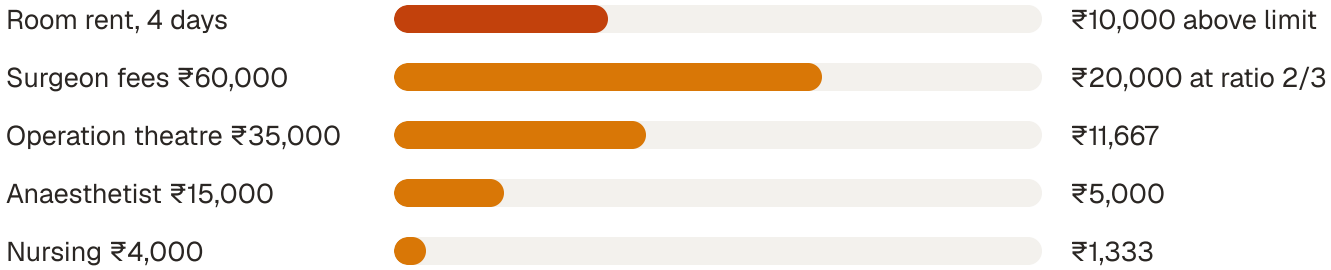
01	**Problem**	The policy's cost shows up at the hospital
02	**Solution**	Coverage Twin: change your room, see the difference, with the clause behind every number
03	**Proof**	A real wording, quoted verbatim, and what the engine refuses to guess
04	**Technology**	A deterministic engine; AI only where a person confirms
05	**USP**	Item-level, clause-cited, honest about unknowns
06	**Impact & benefits**	The room decision made before admission; results measured, not claimed
07	**Business model**	Where it fits Paytm, and the tests before anyone believes a number

Every entry above is a link. Each slide has a Contents link in its footer. Source links in this document open the repository and the regulator's copy of the policy wording.

The policy's cost shows up at the hospital

Room-rent limits, proportionate deductions, sub-limits and exclusions surface at admission, not at purchase. Pick a room above the eligible rate and many policies pay the surgeon, the anaesthetist, nursing and the operation theatre only in the proportion **eligible ÷ actual rate**. It is not about the room.

One room choice, four items that are not the room



Worked example on a synthetic policy and estimate, computed by the prototype. Not a market statistic.

₹2,500 a day

the room-rent gap: ₹7,500 chosen against ₹5,000 eligible

₹48,000

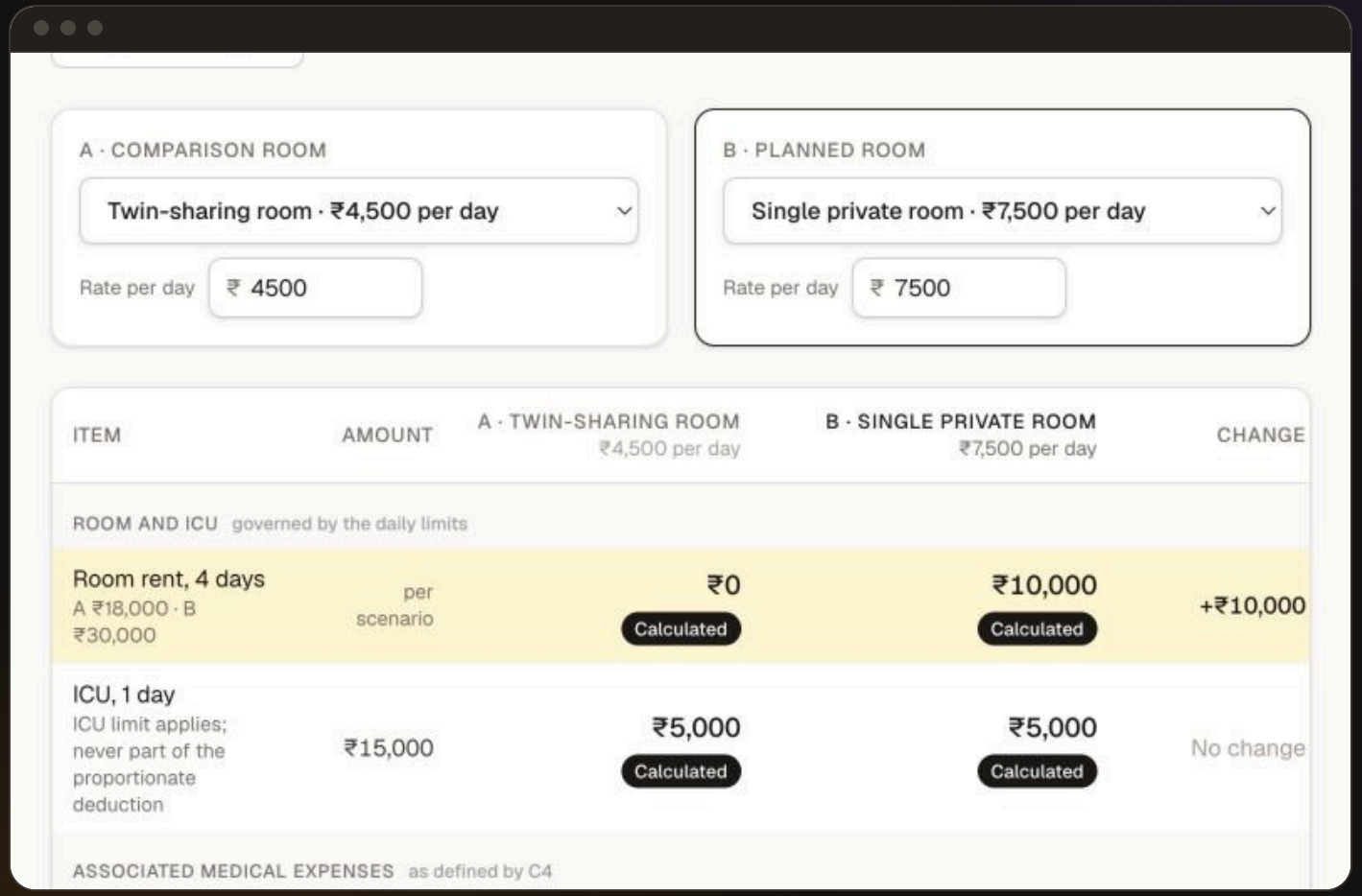
potential deductions from that one choice; ₹38,000 of it on items that are not the room

4 of 11 lines

affected under the policy's definition; 3 excluded by it; 2 the wording never classifies

Which items are touched depends on the policy's own definition. General explainers stop at the concept. The gap we are testing is applying *your* wording to *your* bill.

Change your room. See the difference.



1. **Two rooms, one policy, one estimate**
   Twin-sharing at ₹4,500 against single private at ₹7,500. The amber rows are what changed, and only those.

2. **Every amount opens its evidence**
   Clause and page, inputs with where they came from, the formula, the result. Tampered traces fail validation.

3. **Ranges, not guesses**
   Until the hospital confirms differential billing, the surgeon's deduction is ₹0 to ₹20,000. Confirm it and it becomes ₹20,000, with "confirmed by you" on the trace.

4. **Unknown is never zero**
   Lines the wording never classifies stay out of every total, and the panel says who can settle them.

Documents in, itemised deductions out, every step on screen

1 **Policy documents**
Wording and schedule, synthetic or real, labelled which.

2 **Structured rules**
Each rule cites its clause and quotes it in full.

3 **Scenario simulation**
Room, rate, days, the hospital's own line items.

4 **Itemised potential deductions**
Room excess, ICU excess, proportion, co-payment.

5 **Source clause**
Page-cited, verbatim, beside every number.

6 **Explicit uncertainty**
What is open, what is at stake, who can settle it.

0:00
Overview
₹15,000 calculated, ₹0 to ₹38,000 conditional, ₹7,500 not classified. Not a total, and it says so.

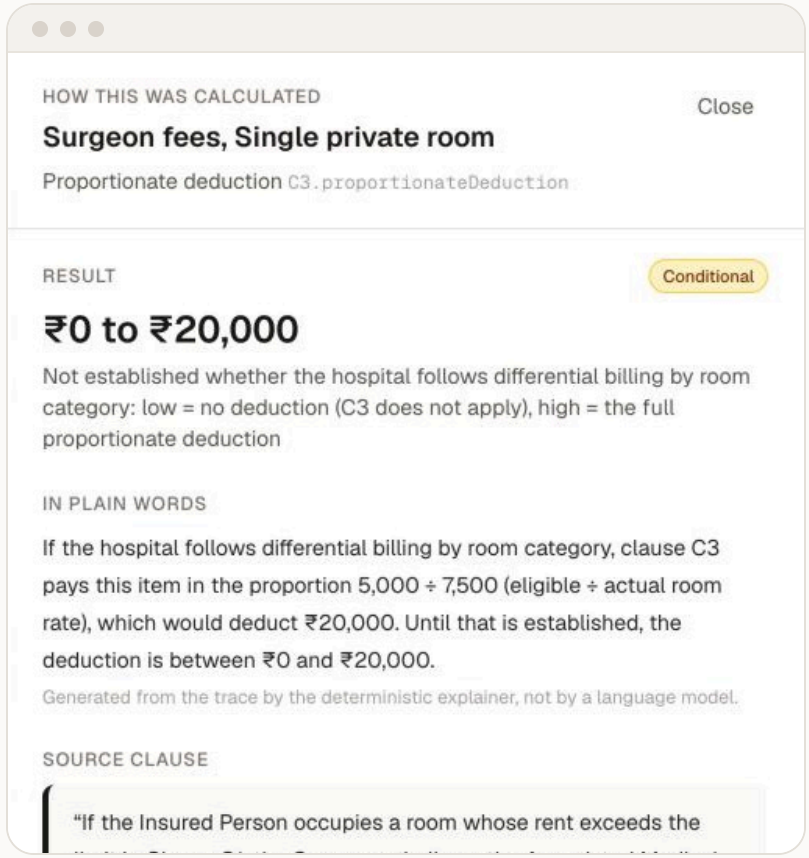
0:25
Twin
Change the room; the itemised breakdown updates at once with the cause rule per row.

1:35
Uncertain
Ask the hospital, ask the insurer, resolve now; plus what this version does not model at all.

3:00
Real wording
Switch to Arogya Sanjeevani and watch the same engine refuse to guess where the wording is silent.

Not a chatbot. Not a payout guarantee. A scenario estimate of the deductions it covers, with its unknowns on screen.

A real wording, quoted verbatim



INSURER-ISSUED · HOSTED BY IRDAI · PARTIALLY ENCODED

Arogya Sanjeevani Policy, The New India Assurance Co. Ltd.

UIN NIAHLIP20165V011920 · the regulator's copy, downloaded 24 October 2026 · 31 pages · ten clauses encoded with page numbers · SHA-256 4e261c78…f542d44 shown on screen

What the engine applies

Room 2% of SI up to ₹5,000 a day; ICU 5% up to ₹10,000; the proportion of all other expenses except medicines when either rate is exceeded; a 5% co-payment.

What it refuses to guess

This 2019-20 wording reduces implants and diagnostics; IRDAI's June 2020 norms later barred that. Those items become a range marked "requires insurer confirmation".

The ICU surprise

A ₹15,000 ICU day alone triggers the proportion under this wording, even in a twin-sharing room. When room and ICU exceed in different ratios, the engine gives bounds, not a number.

Measured

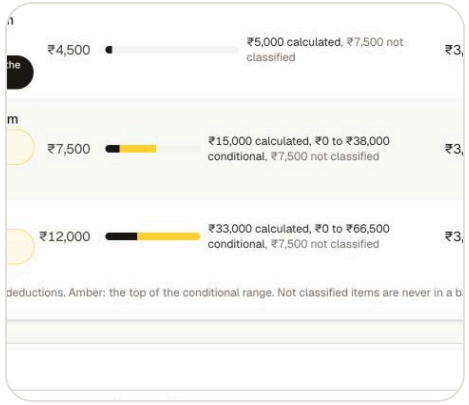
21 of 21 hand-computed amounts exact; 3 of 3 refusal cases right; every trace validated against the verbatim clause text.

Partial encoding, labelled so everywhere. The schedule used with it is synthetic; real schedules are personal and never bundled.

Four tools around one engine

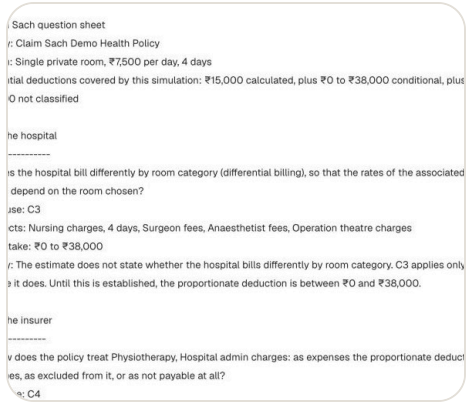
Room ladder

Every room on one view with the deductions at each; the highest room within the limit is marked. One click plans it.



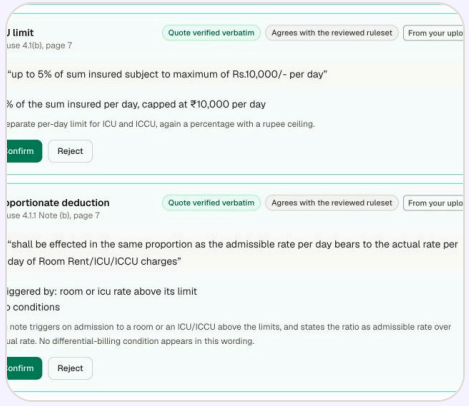
Question sheet

Every open uncertainty becomes a question for the hospital or the insurer, tied to its clause and the rupees at stake. Copy, print or save.



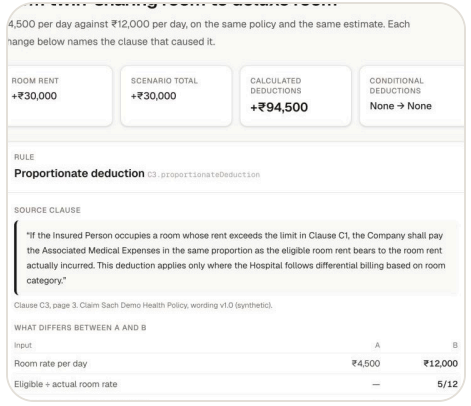
Ruleset builder

A model proposes rules from a public wording with quotes; the engine verifies each quote verbatim; a person confirms. Cached proposals here; the live adapter is off.



Why this changed, and the evidence pack

Each change names its clause, the inputs that differ and both formulas. The whole result exports as a Markdown or JSON evidence pack with every trace.



Deterministic engine, AI only with human review

Frontend

TypeScript, strict. Next.js 16 (App Router), React 19, Tailwind 4. Six screens on a tested screen model; no business logic in components.

Rules engine

Pure TypeScript, no I/O, no model. Integer paise, exact ratios, one half-up rounding. A trace on every amount: rule, inputs, clause, formula, output, open assumptions.

Two wordings

A synthetic demo policy, and Arogya Sanjeevani (New India, from irdai.gov.in): ten clauses quoted verbatim, page-cited, SHA-256 shown. Partially encoded and labelled so.

AI, in its place

The Ruleset builder: a model proposes rules with quotes; the engine verifies each quote verbatim; a person confirms. Cached in this build; live adapter off. Never in the calculation path.

Quality

Vitest, 180 tests, written first. GitHub Actions on every PR. Evaluation harness: six measures over two wordings, results committed and checked for staleness.

Runs anywhere

One command, offline on a laptop; fonts bundled; Vercel-ready. Nothing the user enters leaves the browser. No key, no account, no analytics, no database.

DATA FLOW, ALL IN THE BROWSER

```
fixture or upload (JSON, CSV) → ingestion: schema + verbatim evidence check → screen model → rules engine: classify → limits → proportion → co-payment → traces → six screens, question sheet, evidence pack
```

AI seam, off by default: public clause text → model proposals → verbatim quote check → human confirmation → ruleset. No schedule, bill or name ever leaves the browser.

Item-level, clause-cited, honest about unknowns

1 **Every number is clickable**

Clause and page, inputs with where they came from, the formula, the operation. Tampered traces fail validation.

2 **Unknown is never zero**

Items the wording does not classify stay out of every total. A conditional amount shows both bounds and the condition.

3 **Change your room, see the difference**

Two rooms side by side, the rule behind every changed row; the ladder and the question sheet turn it into a decision and a call script.

4 **Proven on a real wording, honestly**

Arogya Sanjeevani, ten clauses verbatim with a checksum; where the 2019-20 text conflicts with IRDAI's 2020 norms, the engine says so.

What we claim

Claim Sach turns a policy wording into a verifiable hospital-bill simulation: change the room, inspect every modelled deduction, and see what the documents cannot determine.

What we do not claim

Not the first to read clauses; not a payout guarantee; no accuracy figure beyond the measured ones. A bounded review of nine public sources found no tool doing this; we claim we looked, not that nobody does.

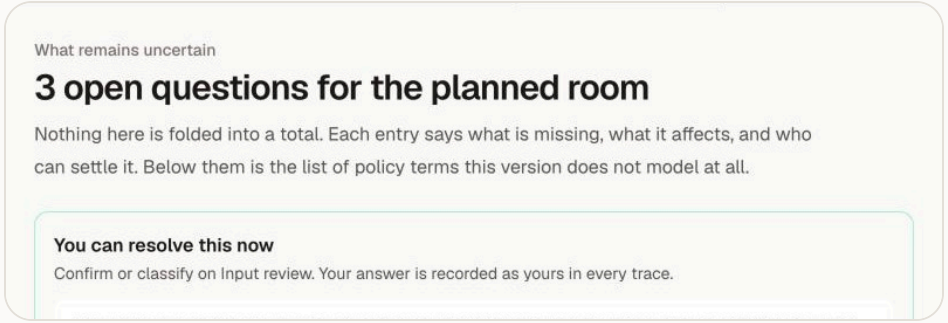
The room decision, made before admission

For the policyholder

The room decision with its deductions visible before admission, not after. Questions for the hospital and the insurer written out: what to ask, and what it is worth.

For a broker's claim-assistance agent

The clause on screen during the call. The uncertainty list is the call script; the evidence pack is the written answer.



MEASURED, NOT CLAIMED

MEASURE	SYNTHETIC	REAL WORDING
Clause retrieval, top-1	11 of 12	10 of 12
Rule extraction, field-level	10 of 10	10 of 12
Calculations, exact paise	9 of 9 · 46 amounts	3 of 3 · 21 amounts
Uncertainty and refusal	6 of 6	3 of 3
Model proposals: quotes verbatim (cached)	n/a	6 of 6
Model proposals: agree with reviewer (cached)	n/a	11 of 13

Measured on the synthetic wording and on ten encoded clauses of one real wording; the model proposals are cached, not a live run. The product shows no accuracy figure anywhere. Figures on this slide are potential deductions covered by the simulation, never total out-of-pocket cost.

Where it fits Paytm, and what must be tested

VERIFIED FACTS

- Paytm Insurance Broking Pvt Ltd is a Direct (Life & General) broker: IRDAI Reg. No. 700, renewed 9 Feb 2026, valid to 16 Feb 2029.
- It distributes car, two-wheeler, health, life and term policies from 18+ insurers, including Niva Bupa and Aditya Birla Health. It does not underwrite.
- It already offers claim assistance by phone. Arogya Sanjeevani is IRDAI's standard product on one wording; one verified encoding serves every insurer that sells it.
- FY25 turnover ₹37.1 crore, loss ₹13.2 crore (Form AOC-1), inside a ₹2,593 crore financial-services line in FY26. Insurance is a small line for Paytm today; clarity is a wedge worth testing.

HYPOTHESES, EACH WITH ITS TEST

HYPOTHESIS	TEST
Fewer claim-time support contacts and disputes	Four-week pilot at one claim-assistance desk, against a matched cohort
More eligible-room choices when the consequence is visible	Room-change rate in the simulator, then claims data
Panel insurers co-fund a coverage-clarity tool	Three conversations; one signed pilot letter is the pass
A coverage check lifts conversion or renewal on the policy page	A/B test, if Paytm chooses to run one

Revenue options depend on those tests: a broker-side feature (retention), an insurer-paid integration, or a premium claim-assistance tier. No Paytm integration is claimed or built. Bima Sugam is not live; it belongs in the future tense.

Built, tested, runs with the Wi-Fi off

Shortlist us and we bring a laptop. Three minutes: change the room, open a number, watch the engine refuse to guess on a real wording.

- A second real wording through the builder, with the live adapter on.
- Independent review of the evaluation fixtures.
- A four-week pilot at a broker's claim-assistance desk.

Code, tests and documents: github.com/Pushkarmondal/paytm-hackathon

